

## Appendix – C: Important Summary and Formulae

**Table 1: Important formulas on probability**

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                     |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| 1. | Mathematical or classical or a priori probability                                                                                                                                                                                                                                                                                                                                                                                                                     | $p = P(E) = \frac{\text{Favourable number of events}}{\text{Total number of events}} = \frac{m}{n}$ |
| 2. | Statistical or empirical or posteriori Probability                                                                                                                                                                                                                                                                                                                                                                                                                    | $p = P(E) = \lim_{n \rightarrow \infty} \frac{m}{n}$                                                |
| 3. | Axioms probability<br>(i) For each $A \in S$ , $P(E)$ is defined, is real and $0 \leq P(E) \leq 1$ .<br>(ii) $P(S) = 1$<br>(iii) If $\{E_n\}$ is any infinite or finite sequence of disjoint events in $S$ , then<br>$P\left(\bigcup_{j=1}^n E_j\right) = \sum_{i=1}^n P(E_i)$<br>(iv) If $\{E_n\}$ is any infinite sequence of disjoint events in $S$ then of disjoint events in $S$ then<br>$P\left(\bigcup_{j=1}^{\infty} E_j\right) = \sum_{i=1}^{\infty} P(E_i)$ |                                                                                                     |
| 4. | Additive law of probability (non Mutually exclusive events)                                                                                                                                                                                                                                                                                                                                                                                                           | $P(A \cup B) = P(A) + P(B) - P(A \cap B)$                                                           |
| 5. | Non mutually exclusive and Independent events                                                                                                                                                                                                                                                                                                                                                                                                                         | $P(A \cup B) = 1 - P(\bar{A}) P(\bar{B})$                                                           |
| 6. | Additive law of probability (mutually exclusive events)                                                                                                                                                                                                                                                                                                                                                                                                               | $P(A \cup B) = P(A) + P(B)$                                                                         |
| 7. | Multiplicative law of probability (Conditional events)                                                                                                                                                                                                                                                                                                                                                                                                                | $P(A \cap B) = P(A) P(B A), P(A) > 0$<br>$= P(B) P(A B), P(B) > 0.$                                 |
| 8. | Multiple law of probability (independent events)                                                                                                                                                                                                                                                                                                                                                                                                                      | $P(A \cap B) = P(A) P(B)$                                                                           |
| 9. | Bayes' theorem                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $P(E_i   A) = \frac{P(E_i) P(A   E_i)}{\sum_{i=1}^n P(E_i) P(A   E_i)} \quad i = 1, \dots, n$       |

**Table 2: Important formulas on Distribution**

|    |                                                                                      |                                                                                                                                                                                              |
|----|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. | Properties of a probability mass function (pmf) for a discrete random variable.      | (i) $P(x) \geq 0$<br>(ii) $\sum_{x=-\infty}^{\infty} P(x) = 1$<br>(iii) $P\left(\bigcup_{j=1}^n E_j\right) = \sum_{i=1}^n P(E_i)$ , if $E_1, E_2, \dots, E_n$ are mutually exclusive events. |
| 2. | Properties of a probability density function (pdf) for a continuous random variable. | (i) $f(x) \geq 0$ for all $x \in R$<br>(ii) $\int_{-\infty}^{\infty} f(x) dx = 1$<br>(iii) $P(a < X < b) = \int_a^b f(x) dx$                                                                 |
| 3. | Cumulative distribution function for a discrete random variable.                     | (i) $F(x) = P(X \leq x) = \sum_{x_i = -\infty}^x P(X = x_i)$ [discrete]<br>(ii) $F(x) = P(X \leq x) = \int_{-\infty}^x f(x) dx$ [continuous]                                                 |
| 4. | Properties of cumulative                                                             | (i) $P(a \leq X \leq b) = F(b) - F(a)$                                                                                                                                                       |

|                       |                                                                                        |
|-----------------------|----------------------------------------------------------------------------------------|
| distribution function | (ii) $F(b) \geq F(a)$ for all $b \geq a$<br>(iii) $F(-\infty) = 0$ and $F(\infty) = 1$ |
|-----------------------|----------------------------------------------------------------------------------------|

Table 3: Important formulas on Expectation and Variance

|                              |                                                                                                                                                                                                                                       |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Expectation               | $E(X) = \sum_{x=-\infty}^{\infty} xP(X=x)$ if $X$ is discrete random variable.<br>$E(X) = \int_{-\infty}^{\infty} xf(x)dx$ if $X$ is continuous random variable.<br>$E(X)$ is denoted as $\mu_X$ or simply $\mu$ to represent mean.   |
| 2. Properties of expectation | (i) $E(aX + b) = aE(X) + E(b) = aE(X) + b$<br>(ii) $E[g(X)] = \sum_{x=-\infty}^{\infty} g(x)P(X=x)$ if $X$ is discrete random variable.<br>(iii) $E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x)dx$ if $X$ is continuous random variable. |
| 3. Variance                  | $V(X) = \sigma_X^2 = E[X - E(X)]^2 = E(X^2) - [E(X)]^2$<br>$V(X) = \frac{1}{n} \sum_{i=1}^n [x_i - E(X)]^2$ for $n$ observations.<br>$V(aX) = a^2 V(X)$                                                                               |

Table 4: Important formulas on discrete distributions

| Name of the distribution | Probability mass function                                                 | Mean                                     | Variance                                     |
|--------------------------|---------------------------------------------------------------------------|------------------------------------------|----------------------------------------------|
| 1. Binomial              | $P(X=x) = {}^nC_x p^x q^{n-x}$ ,<br>$x = 0, 1, 2, \dots, n$ ; $p + q = 1$ | $np$                                     | $Npq$                                        |
| 2. Poisson               | $P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}$ ,<br>$x = 0, 1, 2, \dots$    | $\lambda$                                | $\lambda$                                    |
| 3. Geometric (1)         | $P(X=x) = q^{x-1} p$ ,<br>$x = 1, 2, \dots$ ; $p + q = 1$                 | $\frac{1}{p}$                            | $\frac{q}{p^2}$                              |
| 4. Geometric (2)         | $P(X=x) = q^x p$ ,<br>$x = 0, 1, 2, \dots$ ; $p + q = 1$                  | $\frac{q}{p}$                            | $\frac{q}{p^2}$                              |
| 5. Uniform (discrete)    | $P(X=x) = \frac{1}{n}$ ; $x = x_1, x_2, \dots, x_n$                       | $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ | $\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$ |

Table 5: Important formula on continuous distributions

| Name of the distribution                                 | Probability density function                                                                                         | Mean                | Variance              |
|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------|
| 1. Uniform                                               | $f(x) = \frac{1}{b-a}$ , $a < x < b$                                                                                 | $\frac{b+a}{2}$     | $\frac{(b-a)^2}{12}$  |
| 2. Exponential ( $\lambda$ is the number of occurrences) | $f(x) = \lambda e^{-\lambda x}$ , $0 < x < \infty$ ,<br>$\lambda > 0$ is number of occurrence.                       | $\frac{1}{\lambda}$ | $\frac{1}{\lambda^2}$ |
| 3. Exponential ( $\lambda$ is time per occurrence)       | $f(x) = \frac{1}{\lambda} e^{-x/\lambda}$ , $0 < x < \infty$                                                         | $\lambda$           | $\lambda^2$           |
| 4. Gamma (two parameters)                                | $f(x) = \frac{1}{\Gamma(\alpha) \beta^\alpha} e^{-x/\beta} x^{\alpha-1}$ ,<br>$x > 0$ , $\alpha > 0$ , $\beta > 0$ . | $\alpha\beta$       | $\alpha\beta^2$       |
| 5. Gamma (Single or one Parameter) $\beta = 1$           | $f(x) = \frac{1}{(\alpha-1)!} e^{-x} x^{\alpha-1}$ , $x > 0$ , $\alpha > 0$ .                                        | $\alpha$            | $\alpha$              |

|                    |                                                                                                                                 |       |          |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------|-------|----------|
| 6. Normal          | $f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]$<br>$-\infty < x < \infty$ | $\mu$ | $\sigma$ |
| 7. Standard normal | $f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$ , $-\infty < z < \infty$ .                                                            | 0     | 1        |

Table 6: Important formulas on two-dimensional random variables

|                                                                                 |                                                                                                                                                                                                                                                  |
|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Properties of joint probability mass function of random variable $X$ and $Y$ | (i) $0 \leq P(X=x, Y=y) \leq 1$ for all $X=x$ and $Y=y$<br>(ii) $\sum_x \sum_y P(X=x, Y=y) = 1$                                                                                                                                                  |
| 2. Marginal probability of random variable $X$                                  | $f_X(x) = p_i = \sum_{j=1}^n P(X=x_i, Y=y_j) = P(X=x_i)$ for $i = 1, 2, \dots, m$ ; and $j = 1, 2, \dots, n$ .                                                                                                                                   |
| 3. Marginal probability of random variable $Y$                                  | $f_Y(y) = p_j = \sum_{i=1}^m P(X=x_i, Y=y_j) = P(Y=y_j)$ for $i = 1, 2, \dots, m$ ; and $j = 1, 2, \dots, n$ .                                                                                                                                   |
| 4. Conditional probability of $Y$ give $X=x$                                    | $f_{Y X}(y) = P(Y=y_j   X=x_i) = \frac{P(X=x_i, Y=y_j)}{P(X=x_i)}$ for $i = 1, 2, \dots, m$ and $j = 1, 2, \dots, n$ .                                                                                                                           |
| 5. Cumulative probability Distribution                                          | $P(X \leq x_i, Y \leq y_j) = \sum_{x=-\infty}^x \sum_{y=-\infty}^y P(X=x, Y=y)$                                                                                                                                                                  |
| 6. Joint probability density function of $X$ and $Y$                            | (i) $f(x, y) \geq 0$<br>(ii) $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$<br>(iii) For a given region $R = \{a < X < b \text{ and } c < Y < d\}$ , we have<br>$P(a < X < b, c < Y < d) = \int_a^b \int_c^d f(x, y) dy dx$ |
| 7. Marginal probability density function for $X$                                | $f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$ for $-\infty < x < \infty$ .                                                                                                                                                                       |
| 8. Marginal probability density function for $Y$                                | $f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$ for $-\infty < y < \infty$ .                                                                                                                                                                       |
| 9. Conditional pdf of $X$ given $Y$                                             | $f_X Y(x) = \frac{f(x, y)}{f_Y(y)}$ for the conditional range of $x$                                                                                                                                                                             |
| 10. Conditional pdf of $Y$ given $X$                                            | $f_Y X(y) = \frac{f(x, y)}{f_X(x)}$ for the conditional range of $y$                                                                                                                                                                             |
| 11. Cumulative probability distribution                                         | $F(x, y) = P(-\infty \leq X \leq x, -\infty \leq Y \leq y)$<br>$= \int_{-\infty}^x \int_{-\infty}^y f(x, y) dy dx$                                                                                                                               |
| 12. Independence of $X$ and $Y$                                                 | (i) $P(X=x_i, Y=y_j) = P(X=x_i)P(Y=y_j)$ for all $i$ and $j$ , if $X$ and $Y$ are discrete random variables.<br>(ii) $f(x, y) = f_X(x)f_Y(y)$ , if $X$ and $Y$ are continuous random variables.                                                  |
| 13. Expectation of $XY$                                                         | $E(XY) = \sum_i \sum_j P(X=x_i, Y=y_j)$ , if $X$ and $Y$ are discrete random variables.<br>$E(XY) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f(x, y) dx dy$ , if $X$ and $Y$ are                                                       |



### Probability and Statistics For Engineers

|     |                                           | continuous random variables.                                                                                                         |
|-----|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| 14. | Covariance of $X$ and $Y$                 | $\sigma_{XY} = E(XY) - E(X)E(Y)$                                                                                                     |
| 15. | Conditional expectation (discrete case)   | $E(X   Y = y_j) = \sum_i x_i P(X = x_i   Y = y_j)$ for all $j$ .<br>$E(Y   X = x_i) = \sum_j y_j P(Y = y_j   X = x_i)$ for all $i$ . |
| 16. | Conditional expectation (Continuous case) | $E(X   Y) = \int x f(x   y) dx$<br>$E(Y   X) = \int y f(y   x) dy$                                                                   |

**Table 7: Important definitions and formulas on point estimation**

|    |                                 |                                                                                                                 |
|----|---------------------------------|-----------------------------------------------------------------------------------------------------------------|
| 1. | Methods of estimation           | (i) Point estimation.<br>(ii) Interval estimation.                                                              |
| 2. | Properties of a point estimator | (i) Unbiasedness,<br>(ii) Efficiency,<br>(iii) Consistency,<br>(iv) Sufficiency.                                |
| 3. | Unbiased estimator              | $E(\hat{\theta}) = \theta$ , $\hat{\theta} = \hat{\theta}(x_1, x_2, \dots, x_n)$                                |
| 4. | Simple mean                     | $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$<br>(unbiased estimator of population mean $\mu$ )                      |
| 5. | Sample variance                 | $S^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$<br>(unbiased estimator of population variance $\sigma^2$ ) |

**Table 8: Important formulas on sampling distribution on interval estimation**

|    |                                                                                                                      |                                                                                                                                                                                                                                                                           |
|----|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. | Distribution of the sample mean                                                                                      | $E(\bar{x}) = \mu$ and $V(\bar{x}) = \sigma^2/n$                                                                                                                                                                                                                          |
| 2. | Chi-square Distribution                                                                                              | $f(\chi^2) = \frac{1}{2^{n/2} \Gamma(n/2)} e^{-\chi^2/2} (\chi^2)^{(n/2)-1}$ , $0 < \chi^2 < \infty$<br>$E(\chi^2) = n$ and $V(\chi^2) = 2n$<br>$E(\chi^2) = v = n-1$ and $V(\chi^2) = 2v = 2(n-1)$ , if population mean $\mu$ is estimated from the sample of size $n$ . |
| 3. | Confidence interval for single mean, when $\sigma$ known                                                             | $\left( \bar{x} - \frac{\sigma}{\sqrt{n}}, \bar{x} + \frac{\sigma}{\sqrt{n}} \right)$                                                                                                                                                                                     |
| 4. | Confidence interval for single mean when $\sigma$ unknown                                                            | $\left( \bar{x} - \frac{S}{\sqrt{n}} t_{\alpha/2, n-1}, \bar{x} + \frac{S}{\sqrt{n}} t_{\alpha/2, n-1} \right)$                                                                                                                                                           |
| 5. | Confident interval for difference of two population means, when $\sigma_1, \sigma_2$ known                           | $\left( (\bar{x} - \bar{y}) - z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}, (\bar{x} - \bar{y}) + z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \right)$                                                                     |
| 6. | Confidence interval for difference of two population means, when standard deviations are not known but assumed equal | $\left\{ (\bar{x} - \bar{y}) - t_{\alpha/2, n_1+n_2-2} S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}, (\bar{x} - \bar{y}) + t_{\alpha/2, n_1+n_2-2} S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \right\}$                                                                         |

### Important Summary and Formulae

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | where $S_p = \sqrt{\frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2}}$                                                                                                                                                                                                                                                                                                                                                                                         |
| 7.  | Confidence interval for difference of two population means when standard deviations are not known and not equal<br>$\left( (\bar{x} - \bar{y}) - t_{\alpha/2} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}, (\bar{x} - \bar{y}) + t_{\alpha/2} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}} \right)$<br>where $v = \frac{\left( \frac{S_1^2}{n_1} + \frac{S_2^2}{n_2} \right)^2}{\left( \frac{S_1^2}{n_1} \right)^2 + \left( \frac{S_2^2}{n_2} \right)^2}$ |
| 8.  | Confidence interval for single population proportion<br>$\left( \hat{p} - z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}, \hat{p} + z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \right)$                                                                                                                                                                                                                                                            |
| 9.  | Confidence interval for difference of two population proportions<br>$\left( (\hat{p}_1 - \hat{p}_2) - z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}, (\hat{p}_1 - \hat{p}_2) + z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}} \right)$                                                                                                                          |
| 10. | Sample size<br>$n = \left( \frac{\sigma}{E} z_{\alpha/2} \right)^2$ , where $\sigma$ is known<br>$n = \left( \frac{S}{E} t_{\alpha/2} \right)^2$ , where $\sigma$ is unknown                                                                                                                                                                                                                                                                               |
| 11. | Error<br>$E = \frac{\sigma}{\sqrt{n}} z_{\alpha/2}$ or $E = \frac{S}{\sqrt{n}} z_{\alpha/2}$                                                                                                                                                                                                                                                                                                                                                               |

**Table 9: Important formulas on testing of hypothesis**

|    |                                                                                                                                                                                        |                                                                                                                                                                       |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. | (i) P(Type I Error)<br>(ii) P(Type II Error)                                                                                                                                           | $P(\text{Reject } H_0   H_0 \text{ is true}) = \alpha$<br>$P(\text{Accept } H_0   H_0 \text{ is false}) = \beta$                                                      |
| 2. | Power of the test                                                                                                                                                                      | $P(\text{Reject } H_0   H_0 \text{ is false}) = 1 - \beta$                                                                                                            |
| 3. | Test statistic                                                                                                                                                                         | $\frac{\text{Sample statistic} - E(\text{Sample statistic})}{\text{Standard error of sample statistic}}$                                                              |
| 4. | Test statistic for single mean<br>$H_0: \mu = \mu_0$<br>$H_1: \mu \neq \mu_0$ (two-tail test)<br>$H_1: \mu < \mu_0$ , or<br>$H_1: \mu > \mu_0$ (one tailed test)                       | $Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$ (Normal test: large sample and population variance known)                                                               |
| 5. | Test statistic for the difference of two means<br>$H_0: \mu_1 = \mu_2$<br>$H_1: \mu_1 \neq \mu_2$ (two-tail test)<br>$H_1: \mu_1 < \mu_2$ or<br>$H_1: \mu_1 > \mu_2$ (one tailed test) | $Z = \frac{(\bar{x} - \bar{y}) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$ (Normal test: large samples and population variance known) |
| 6. | Test statistic for single proportion<br>$H_0: p = p_0$<br>$H_1: p \neq p_0$ (two-tail test)<br>$H_1: p < p_0$ , or<br>$H_1: p > p_0$                                                   | $Z = \frac{\hat{p} - p}{\sqrt{p(1-p)/n}}$ (Normal test: large sample)                                                                                                 |

END

# Probability and Statistics For Engineers

|     |                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                       |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7.  | $H_0: p_1 > p_0$ (one tailed test)<br>Test statistic for the difference of two proportions<br>$H_0: p_1 = p_2$<br>$H_1: p_1 \neq p_2$ (two-tail test), or<br>$H_1: p_1 < p_2$ , or<br>$H_1: p_1 > p_2$ (one tailed test) | $Z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}}$ or $Z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$ where $\hat{p} = \frac{n_1\hat{p}_1 + n_2\hat{p}_2}{n_1 + n_2}$<br>(Normal test: large samples) |
| 8.  | Test statistic for single mean<br>$H_0: \mu = \mu_0$<br>$H_1: \mu \neq \mu_0$ (two-tail test)<br>or, $H_1: \mu < \mu_0$<br>or, $H_1: \mu > \mu_0$ (one tail tests)                                                       | $t = \frac{\bar{x} - \mu}{S/\sqrt{n}}$ (t-Test: small sample, population variance unknown)                                                                                                                                                                                                                                            |
| 9.  | Test statistic for the difference of two means<br>$H_0: \mu_1 = \mu_2$<br>$H_1: \mu_1 \neq \mu_2$ (two-tail test)<br>or, $H_1: \mu_1 < \mu_2$<br>or, $H_1: \mu_1 > \mu_2$ (one tail tests)                               | $t = \frac{(\bar{x} - \bar{y}) - (\mu_1 - \mu_2)}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$ , where<br>$S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1 + n_2 - 2}$<br>(t-Test: small samples, population variances unknown but equal)                                                                                                |
| 10. | Test statistic for the difference of two means<br>$H_0: \mu_1 = \mu_2$<br>$H_1: \mu_1 \neq \mu_2$ (two-tail test)<br>Or, $H_1: \mu_1 < \mu_2$<br>Or, $H_1: \mu_1 > \mu_2$ (one tail tests)                               | $t = \frac{(\bar{x} - \bar{y}) - (\mu_1 - \mu_2)}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$<br>where degree of freedom is given as<br>$v = \frac{\frac{S_1^2/n_1}{(S_1^2/n_1)^2} + \frac{S_2^2/n_2}{(S_2^2/n_2)^2}}{\frac{1}{n_1-1} + \frac{1}{n_2-1}}$<br>(t-Test: small samples, population variances unknown and unequal)     |
| 11. | Test for equality of two variances<br>$H_0: \sigma_1^2 = \sigma_2^2$<br>$H_1: \sigma_1^2 > \sigma_2^2$ or<br>$H_1: \sigma_1^2 < \sigma_2^2$ (right-tail tests)                                                           | $F = \frac{S_2^2}{S_1^2}$ or $F = \frac{S_1^2}{S_2^2}$ (F: test small samples)                                                                                                                                                                                                                                                        |
| 12. | Test for goodness of fit<br>(i) Independence<br>(ii) Homogeneity<br>(iii) Equality of several proportions                                                                                                                | $\chi^2 = \sum_{i=1}^n \frac{[O - E]^2}{E}$                                                                                                                                                                                                                                                                                           |
| 13. | Correlation between X and Y                                                                                                                                                                                              | $r_{XY} = \frac{\Sigma xy - n\bar{x}\bar{y}}{\sqrt{\Sigma x^2 - n\bar{x}^2} \sqrt{\Sigma y^2 - n\bar{y}^2}}$                                                                                                                                                                                                                          |

## Important Summary and Formulae

|     |                             |                                                                                                                                                          |
|-----|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 14. | Regression line of Y on X   | $y - \bar{y} = b_{yx}(x - \bar{x})$ , where<br>$b_{yx} = \frac{n\Sigma xy - \Sigma x\Sigma y}{n\Sigma x^2 - (\Sigma x)^2} = r \frac{\sigma_y}{\sigma_x}$ |
| 15. | Normal equations for Y on X | $\Sigma y = na + b\Sigma x$ ,<br>$\Sigma xy = a\Sigma x + b\Sigma x^2$                                                                                   |
| 16. | Regression line of X on Y   | $x - \bar{x} = b_{xy}(y - \bar{y})$ , where<br>$b_{xy} = \frac{n\Sigma xy - \Sigma x\Sigma y}{n\Sigma y^2 - (\Sigma y)^2} = r \frac{\sigma_x}{\sigma_y}$ |
| 17. | Normal equations for X on Y | $\Sigma x = nc + d\Sigma y$ ,<br>$\Sigma xy = c\Sigma y + d\Sigma y^2$                                                                                   |
| 18. | Correlation between X and Y | $r_{xy} = \sqrt{b_{yx}b_{xy}}$ , since $b_{yx} = r \frac{\sigma_y}{\sigma_x}$ , $b_{xy} = r \frac{\sigma_x}{\sigma_y}$                                   |